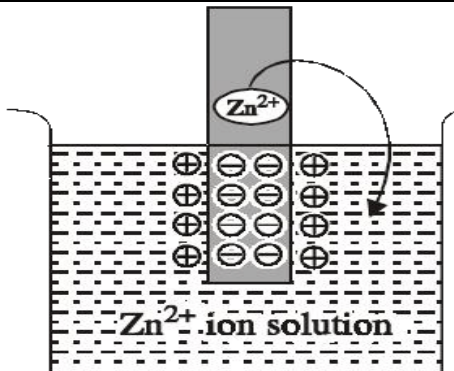
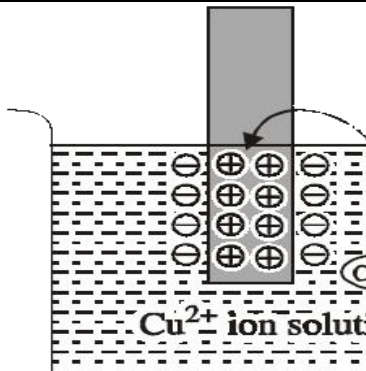
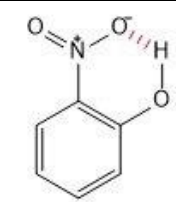
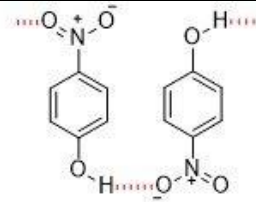
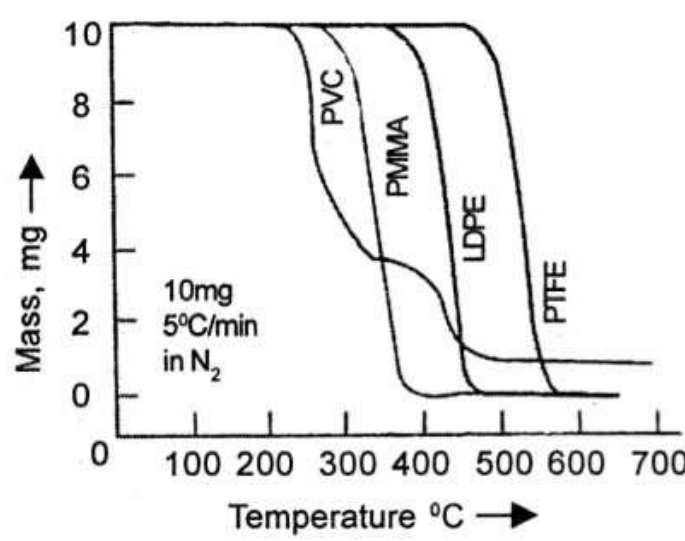
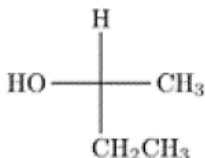
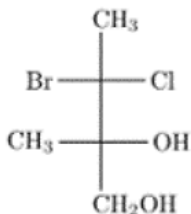
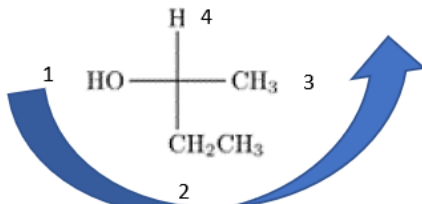
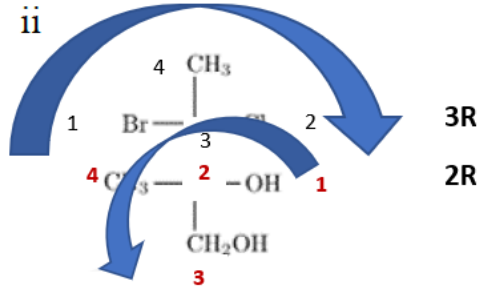
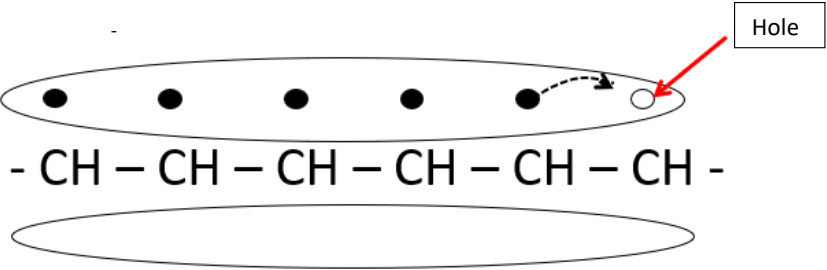
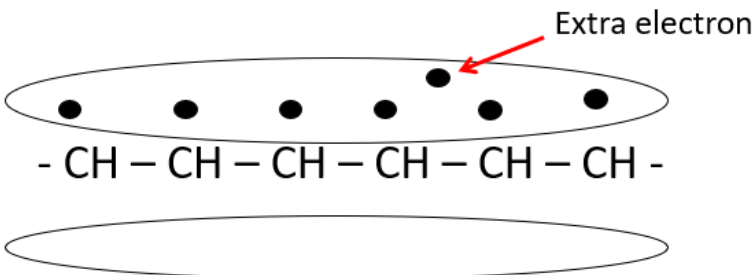


Sahrdaya College of Engineering and Technology, Kodakara		
Answer Key		
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY		
FIRST SEMESTER B.TECHDEGREE EXAMINATION, DECEMBER2021		
Course Code: CYT 100		
Course Name:ENGINEERING CHEMISTRY		
PART A		
		<i>Answer all questions. Each question carries 3 marks</i>
	Q1	How is Helmholtz electrical double layer formed?
	ans	A metal (M) consists of metal ions (M^{n+}) with valence electrons. When the metal (M) is placed in a solution of its own salt, any one of the following reactions will occur. (i) Positive metal ions may pass into the solution. $M \longrightarrow M^{n+} + ne^{-}$ (oxidation) (ii) Positive metal ions from the solution may deposit over the metal. $M^{n+} + ne^{-} \longrightarrow M$ (reduction) Examples – 1 : Zn electrode dipped in $ZnSO_4$ solution When Zn electrode is dipped in $ZnSO_4$ solution, Zn goes into the solution as Zn^{2+} ions. Now, the Zn electrode attains a negative charge, due to the accumulation of valence electrons on the metal. The negative charges developed on the electrode attract the positive ions from solution. Due to this attraction the positive ions remain close to the metal. <i>Example -2: Cu electrode dipped in $CuSO_4$ solution</i> When Cu electrode is dipped in $CuSO_4$ solution, Cu^{2+} ions from the solution deposit over the metal. Now, the Cu electrode attains a positive charge, due to the accumulation of Cu^{2+} ions on the metal. The positive charges developed on the electrode attract the negative ions from solution. Due to this attraction, the negative ions remain close to the metal.

	 1. Zn electrode in ZnSO₄ solution  2. Cu electrode in CuSO₄ solution Thus, a sort of layer (positive (or) negative ions) is formed all around the metal. This layer is called <u>Helmholtz electrical double layer</u>. This layer prevents further passing of the positive ions from or to the metal. A difference of potential is consequently set up between the metal and the solution. At equilibrium, the potential difference becomes a constant value, which is known as the electrode potential of a metal.
Q2	Calculate the single electrode potential of dichromate electrode at 25 °C when [Cr₂O₇²⁻] is 0.3M, [Cr³⁺] is 0.02M and [H⁺] is 1M. Given: Cr₂O₇²⁻ + 14 H⁺ + 6e⁻ → 2Cr³⁺ + 7H₂O ; E₀ = 1.33 V
ans	Cr₂O₇²⁻ + 14 H⁺ + 6e⁻ → 2Cr³⁺ + 7H₂O ; E₀ = 1.33 V E₀ = 1.33 V $E = E_0 - \frac{0.0591}{6} \log \frac{[\text{Cr}^{3+}]^2 \times 1}{[\text{Cr}_2\text{O}_7^{2-}] \times 1^{14}}$ $= 1.33 - \frac{0.0591}{6} \log \frac{0.02^2}{0.3 \times 1^{14}} = 1.358\text{V}$
Q3	Recognize the atoms showing NMR phenomenon among the following. Give Reason. a) ¹H¹ b) ¹H² c) ¹H³ d) ¹⁶O⁸ e) ¹⁸O⁸ f) ¹⁴N⁷
ans	a) ¹H¹ b) ¹H² c) ¹H³ f) ¹⁴N⁷ Only these atoms will show NMR phenomenon because only those nuclei having odd number of protons or odd number of neutrons or both odd number will show NMR phenomenon as they have their spin quantum number greater than 0
Q4	IR spectroscopy can be used to differentiate intra molecular and inter molecular hydrogen bonds. Explain with an example

ans	 <i>o</i>-nitrophenol Intramolecular  <i>p</i>-nitrophenol Intermolecular To distinguish between intermolecular and intramolecular hydrogen bonding, IR spectrum of the compound is recorded at different concentrations In IR spectrum of a compound with intermolecular H-bonding signal of the hydrogen bonding will show variations with dilution. As solvent molecules comes in between the molecules, inter molecular hydrogen bonds become weaker On the other hand, if the hydrogen bonding is intra in nature, there will not be any deviation for the signal of hydrogen bonding during dilution
5	Compare the thermal stability of PVC, PMMA, LDPE and PTFE using TG given below. Justify your answer. 
Ans	PVC is the least thermally stable polymer where as PTFE is the most stable among the given polymer. The thermal stability increases in the order PVC < PMMA < LDPE < PTFE. As the temperature increases PVC undergoes disintegration slightly above 210°C leading to a sudden decrease in its mass. In the case of PTFE no sudden decrease in its mass upto 500°C and the disintegration of the polymer takes place only around 500°C
6	Give the principle of TLC. Mention two applications of TLC.
	TLC is an example of partition chromatography. The stationary phase in this technique is the solvent in which the sample of the mixture of compounds is dissolved and spotted on the chromatographic plate. This solvent along with the sample components will get absorbed on the solid support on the plate and will remain stationary. The mobile phase is the solvent in the glass tank in which the bottom portion of the plate

	is dipped. As the mobile phase moves up the plate it will carry different components in the mixture to different heights on the plate based on their difference in partition coefficient and thereby their separation occurs Applications It is used in the analysis of drug residues and antibiotics It is used in the separation and analysis food samples It is used for separation and identification of components in plant extract in biotechnology
7	Assign the R/S notation to the following compounds i.  ii 
Ans	i.  ii  i) Anti clockwise rotation and Lowest priority group in vertical position so, S. ii) Second chiral carbon shows Anti clockwise rotation and Lowest priority group in Horizontal position so, 2R, whereas third chiral carbon shows clockwise rotation and Lowest priority group in vertical position so, 3R
8	Explain doping in conducting polymers.
Ans	Conductivity of intrinsically conducting polymers is further increased by creating positive or negative charge on the polymer back bone by doping. This positive charge moves throughout the orbital by subsequent jumping of neighbouring electrons and increases the conductivity of the material <u>P- doping (Oxidation)</u> In p-doping, an intrinsically conducting polymer is treated with an oxidising agent

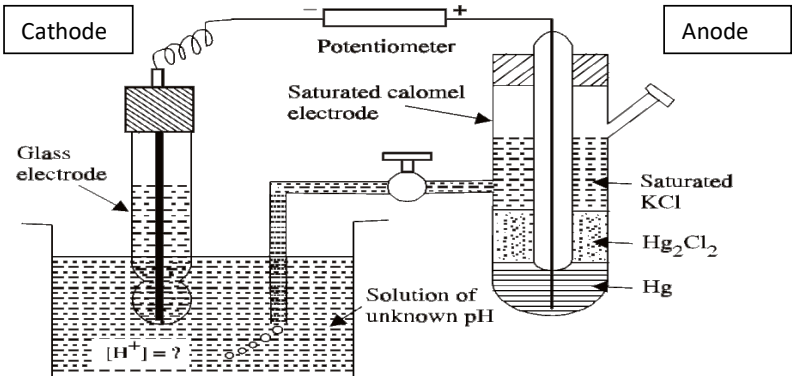
		(Lewis acid or halogens) which removes electron from the polymer and create positively charged sites or <u>holes</u> on the polymer back bone  <u>n-doping(Reduction)</u> • In n-doping, an intrinsically conducting polymer is treated with a reducing agent (Lewis base or alkali metal) which adds electrons to the polymer backbone. E.g for n-dopant- Na, K • These extra electrons move throughout the delocalized π molecular orbital and increases the conductivity of the material 
9		A sample of water on analysis gives the following results. $\text{Ca}^{2+} = 200 \text{ mg/L}$, $\text{Mg}^{2+} = 180 \text{ mg/L}$, $\text{HCO}_3^- = 360 \text{ mg/L}$, $\text{Cl}^- = 200 \text{ mg/L}$ and $\text{Na}^+ = 80 \text{ mg/L}$. Calculate temporary and permanent hardness.
Ans		Total Hardness= $200 \times 100/40 + 180 \times 100/24 = 1250 \text{ ppm}$ Temp Hardness = $360 \times 100/(2 \times 61) = 295 \text{ ppm}$ Perm. Hardness = $1250 - 295 = 955 \text{ ppm}$
10		Differentiate between aerobic and anaerobic oxidation.

	Aerobic process	Anaerobic process
	1. Natural Biological degradation by microbes in presence of oxygen.	Natural biological degradation by microbes in the absence of oxygen.
	2. During oxidation pollutants are broken down CO ₂ , H ₂ O, Nitrates, Phosphates and biomass	broken down to methane gas, CO ₂ , Acetic acid and relatively little amount biomass.
	3. Products are non-combustible.	Biogas is formed.
	4. Products does not have offensive smell.	Most of the products have offensive smell.
	4. Is a fast process.	Is relatively slow.

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

Q11	a) How is glass electrode used in determining the pH of a solution ? What are the advantages and limitations of a glass electrode? (8) b) Describe the principle and applications of Electroless copper plating. (6)
Ans	11a)Determination of pH of a Solution using GlassElectrode The glasselectrodeisplacedinthesolutionundertestandiscoupledw ithsaturatedcalomelelectrodeasshowninthe figure  Fig. 1.6 Determination of pH by using glass electrode Theemfofthecellismeasured.Fromtheemf,thepHofthesolutioniscalculatedasfollows $E_{cell} = E_{cathode} - E_{anode}$ $E_{cell} = E_G - E_{cal}$

	$=E^{\circ}G-0.0591\text{pH}-E_{\text{cal}}$ $=E^{\circ}G-E_{\text{cal}}-0.0591\text{pH}$ $\text{pH} = \frac{E^{\circ}G-E_{\text{cal}}-E_{\text{cell}}}{0.0591}$ Since $E_{\text{cal}}=0.2422\text{V}$ $\text{pH} = \frac{E^{\circ}G-0.2422-E_{\text{cell}}}{0.0591}$ <u>Advantages</u> 1. It can be used to determine the pH of oxidising and reducing agents in presence of proteins which interfere with other electrodes 2. It can be used in the pH range of 0-12 3. It can be used in the coloured, turbid, colloidal solutions containing metal ions 11b) Metal ion + Reducing Agent $\rightarrow$ Metal + Oxidising agent A method of depositing a metal from its salt solution on a catalytically active surface of the metal to be plated using a suitable reducing agent without using electrical energy. It is used to deposit a layer of copper on a substrate without the use of electrical energy. The object to be plated is degreased using organic solvent and this followed by acid treatment. This object is then immersed in plating bath containing cupric salt, reducing agents, buffer and complexing agent <i>Electroless Plating of Copper</i> Components of electroless plating bath of copper 1. Copper sulfate 2. Formaldehyde as reducing agent 3. Caustic basic medium 4. Chelating agents like amines, EDTAs, and tartarates
--	--

		By maintaining a pH of 11-12 using a buffer solution (NaOH 12-15 g/l and Rochelle salt 14 g/l) and at an optimum temperature of around 25°C the following redox reactions takes place. At Anode: $2\text{HCHO} + 4\text{OH}^- \rightarrow 2\text{HCOO}^- + 2\text{H}_2\text{O} + \text{H}_2 + 2\text{e}^-$ At Cathode: $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ Net redox reaction: $2\text{HCHO} + 4\text{OH}^- + \text{Cu}^{2+} \rightarrow 2\text{HCOO}^- + 2\text{H}_2\text{O} + \text{H}_2 + \text{Cu}$
	Q12	a) Write the principle and procedure for the estimation of ferrous ion using dichromate solution potentiometrically. $E_0 \text{ Fe}^{3+}/\text{Fe}^{2+} = 0.77 \text{ V}$ (8) b) Emf of an electrochemical cell is 1.5175 V at 200C and 1.5213 V at 35°C.If the cell reaction involve 2 electrons, find the standard emf of the cell and the reaction quotient. (6)
	ans	12a) Example : Titration between ferrous sulfate and potassium dichromate Take ferrous sulfate in the titration flask with a platinum wire to make electrical contact Couple this with a calomel electrode. Known concentration of Potassium dichromate is added from the burette and the electrode potential is measured after each addition. oxidized state + ne⁻ → reduced state The potential of the indicator electrode is $E = E^\circ + \frac{0.0591}{n} \log \frac{[\text{oxidised state}]}{[\text{reduced state}]}$

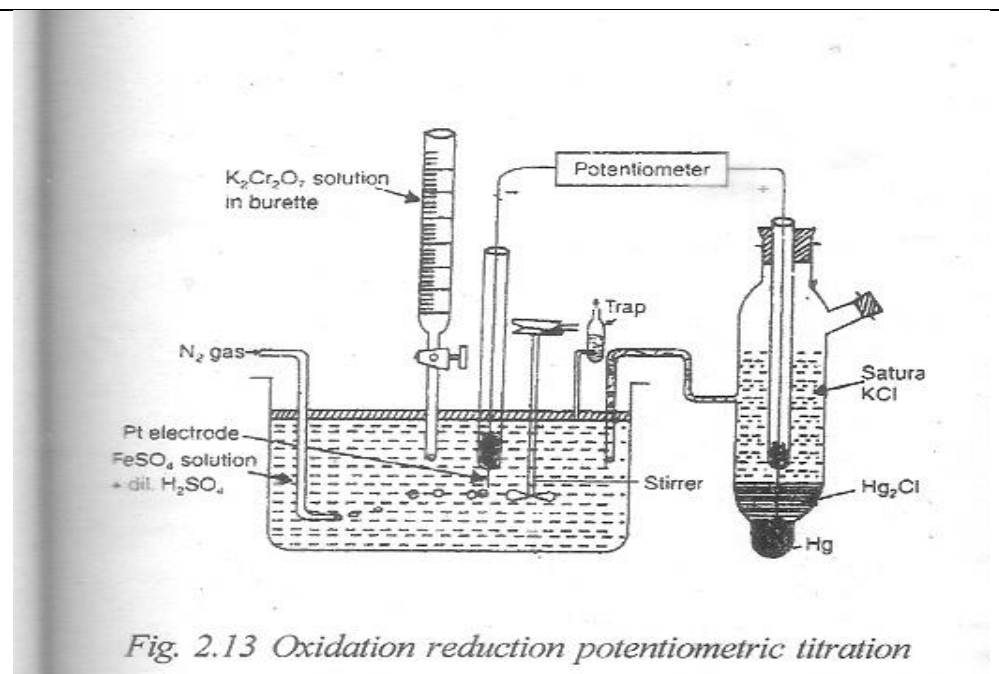


Fig. 2.13 Oxidation reduction potentiometric titration

$$V_1 M_1 = V_2 M_2$$

$$V_{Fe^{2+}} \times M_{Fe^{2+}} = V_{K_2Cr_2O_7} \times M_{K_2Cr_2O_7}$$

$$50 \times M = \text{volume from graph} \times M (\text{known})$$

M can be calculated

12b)

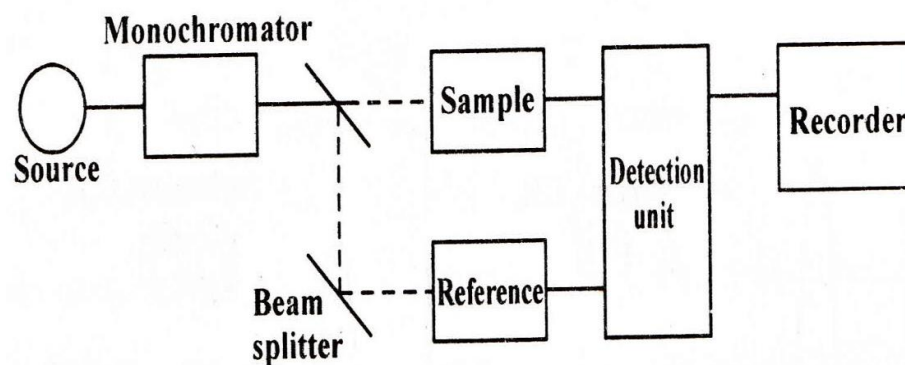
$$1.5175 = E^0_{\text{cell}} - \frac{2.303RT}{nF} \log Q$$

$$1.5175 = A + B \times 293 \text{ ----(1)}$$

$$1.5213 = A + B \times 308 \text{ ----(2)}$$

On solving

	$B = 2.5333 \times 10^{-4}$ $A = 1.4432V = E^0_{\text{cell}}$ $B = -\frac{2.303R}{nF} \log Q$ $2.5333 \times 10^{-4} = -\frac{2.303 \times 8.314}{2 \times 96500} \log Q$ $\log Q = -2.5535$ $Q = 2.795 \times 10^{-3}$
	Module 2
Q13	a) What are the various types of electronic transitions possible in organic molecules? Give examples of each. Also give the instrumentation of UV Visible spectrophotometer. b) Suggest structural formula for the following compounds such that they give a single signal in proton NMR spectroscopy. a) $C_9H_{18}O$ and b) $C_{12}H_{18}$
ans	13 a) <u>$\sigma \rightarrow \sigma^*$ transitions</u> This occur in the compounds with single bonds and no lone pair of electrons Eg. CH_4, $CH_3 - CH_3$, This transitions require high energies / lower wavelength ($< 200 \text{ nm}$) radiations in the UV region <u>$\pi \rightarrow \pi^*$ transitions</u> This occur in compounds with π electron system eg : $CH_2 = CH_2$, Aromatic compounds <u>$n \rightarrow \sigma^*$ transitions</u> This occur in the saturated compounds with heteroatoms and lone pair eg : $(CH_3)_3N$ <u>$n \rightarrow \pi^*$ transition</u> This is shown by unsaturated molecules with hetero atoms like N, O, S. such as aldehyde, ketone etc $\begin{array}{c} O \\ \\ CH_3 - C - CH_3 \end{array}$ $\begin{array}{c} NH_2 \\ \\ CH_3 - CH = CH - C - CH_3 \end{array}$ <u>Instrumentation (UV-Visible Spectrometer)</u>



Components of a UV-Visible spectrometer are

1) Radiation source

A hydrogen or deuterium lamp is used as the radiation source. It should be stable, sufficiently intense and supply continuous radiation

2) Monochromator

Monochromator function is to disperse the radiation according to the wavelength. Essential elements of a monochromator are an entrance slit, a dispersing element usually prism or grating and an exit slit

3) Cells

The cells containing sample or reference should fulfil the following requirements. They must be uniform in construction. The material should be inert to the solvents. They must transmit the light of the used wavelength

4) Detectors

Detectors convert the radiation falling on it into current. Current produced will be inversely proportional to the concentration of the solution. Three common types of detectors used in visible- UV spectrometers are photomultiplier tube, photocell, and Barrier layer cell

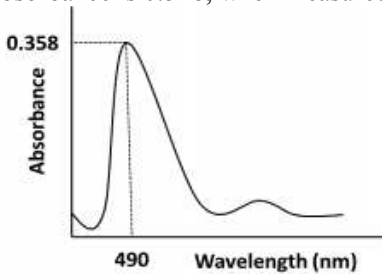
5) Recorder

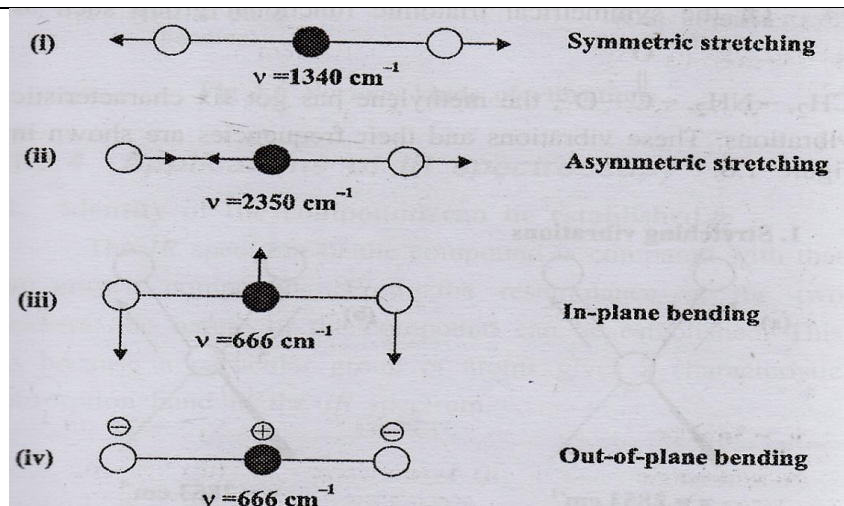
The signal from the detector is finally received by the recorder and the spectrum is drawn by the recorder pen

Working of the spectrophotometer

A beam of radiation from the source passes through the monochromator. The monochromator allows a narrow range of wavelength to come out through an exit slit. This beam is split into two equal halves of same intensity by a beam splitter. One of these beams (sample beam) passes through a transparent sample cell containing the solution of the compound. The other beam (reference beam) passes through an identical cell containing only the solvent. Both the beams fall on the detector and the instrument is designed to compare the intensities of both the beams. When the compound absorbs light at a particular wavelength, the intensity of the sample beam will be less than that of the reference beam. The instrument gives output graph which is a plot of wavelength against absorbance

13b)

	a) $C_9H_{18}O$ $ \begin{array}{ccccccc} & & CH_3 & & O & & CH_3 \\ & & & & & & \\ CH_3 & - & C & - & C & - & C & - & CH_3 \\ & & & & & & \\ & & CH_3 & & & & CH_3 \end{array} $ b) $C_{12}H_{18}$ $ \begin{array}{ccccccc} & & CH_3 & & & & CH_3 \\ & & & & & & \\ CH_3 & - & C & - & C & = & C & = & C & - & C & - & CH_3 \\ & & & & & & & & & & \\ & & CH_3 & & & & & & & & CH_3 \end{array} $
Q14	a) Write the various modes of vibration possible for HCl, CO₂ and H₂O and state which of these modes are IR active. Write reason for their IR activity. b) State Beer- Lambert's law and explain the term molar extinction co-efficient. Given is the absorption spectrum of a compound A of 2.5×10^{-6} M concentration, when measured using 1 cm cuvette in a UV-Vis spectrometer. Calculate the unknown concentration of a test sample of compound A if the absorbance is 0.518, when measured in the same condition.  The graph shows Absorbance on the y-axis and Wavelength (nm) on the x-axis. A prominent peak is observed at 490 nm with an absorbance value of 0.358. There is a smaller, broader peak around 550 nm.
ans	14a) HCl No. of fundamental modes of vibrations = 1 Stretching vibration – IR active CO₂ CO₂ is a linear triatomic molecule and has $(3 \times 3 - 5) = 4$ modes of vibrations These modes and frequencies are shown

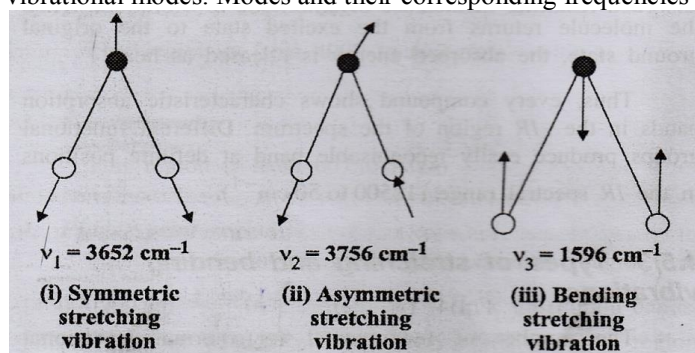


Of the 4 fundamental modes of vibrations, symmetric stretching vibration (i) does not involve any change in dipole moment and hence it is IR inactive

The remaining 3 modes (ii), (iii) and (iv) involve change in dipole moment and hence IR active. Again modes (iii) and (iv) have the same frequency and hence IR spectrum of CO_2 has only two absorption bands

H₂O

Water is a non-linear triatomic molecule and has $3n-6 = (3 \times 3 - 6) = 3$ fundamental vibrational modes. Modes and their corresponding frequencies are shown below



Stretching frequencies are much higher than bending frequencies, ie. More energy is required to stretch a bond than to bend it. All the 3 modes of vibrations are accompanied by change in dipole moment and hence all the 3 vibrations are IR active. Thus IR spectrum of water exhibits 3 absorption bands

14b)

The Beer – Lambert law states that when a beam of monochromatic radiation is passed through a solution of an absorbing substance, the rate of decrease of intensity of radiation with thickness of the absorbing medium is proportional to the intensity of the radiation 'I' as well as the concentration of the absorbing molecules in the solution 'C'

The mathematical expression for the law is

$$A = \epsilon Cx$$

	where $\epsilon = k / 2.303$ and is called molar absorptivity coefficient or molar extinction coefficient. It is a constant for a particular type of absorbing molecules in the solution $A = \log I_0 / I$ is known as absorbance or optical density Application of Beer – Lambert law is determination of unknown concentration of a substance in a solution $A_u / A_s = C_u / C_s$ Therefore $C_u = A_u / A_s \times C_s$ $= 0.518 / 0.358 \times 2.5 \times 10^{-6}$ $= 3.62 \times 10^{-6}$ Test solution concentration = $3.62 \times 10^{-6} \text{ M}$
Module-III	
Q15	a) Sketch the instrumentation of DTA and explain the principle. Explain the DTA of $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$. b) Briefly explain the principle and instrumentation of SEM with the help of a diagram.
ans	15a) Differential Thermal Analysis (DTA) In DTA, the material sample and an inert reference material are heated in a furnace at a constant rate The Differential temperature ($\Delta T = T_s - T_R$), where T_s and T_R are the temperatures of sample and reference respectively, is plotted against temp of the furnace When the sample undergoes a change of state, heat is either absorbed (endothermic) or liberated (exothermic) and this is obtained in the form of a peak DTA Apparatus

Components

It has the same type of furnace, heating system and recorder as employed in TGA. But the furnace has 2 chambers connected with temperature sensors / Thermocouples

Procedure

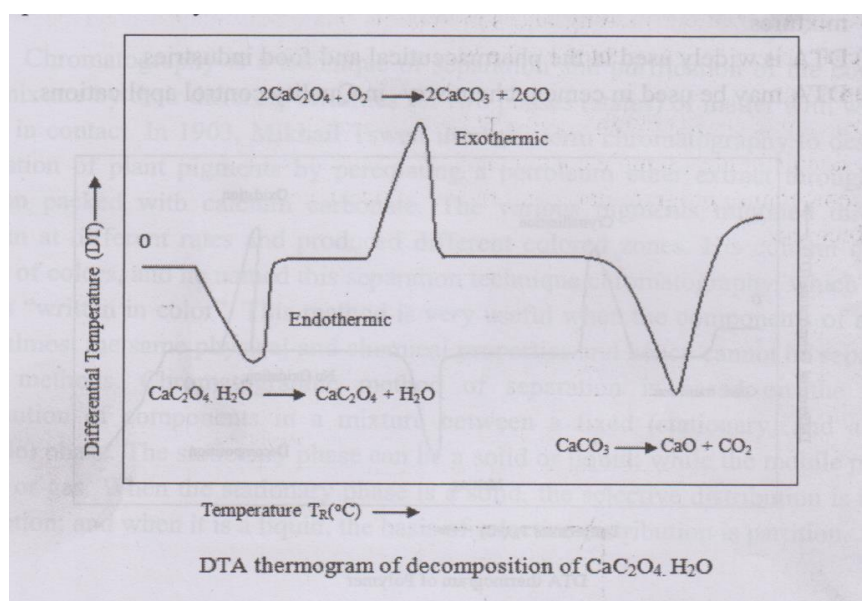
The sample is placed in one chamber and the inert reference material in other chamber.

The furnace is heated by an electrical source

The difference in temperature between the sample(T_S) and the reference (T_R) is continuously measured as a function of the furnace temperature(T)

The DTA thermogram obtained is a plot of ΔT versus reference temperature

DTA thermogram of decomposition of Calcium oxalate



The thermogram contains two minima.

The first minima indicates the dehydration of $\text{CaC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ and the process is endothermic

The maxima indicate decomposition of calcium oxalate into CaCO_3 and CO

The second minima indicates decomposition of CaCO_3

15b)

Scanning Electron Microscope (SEM)

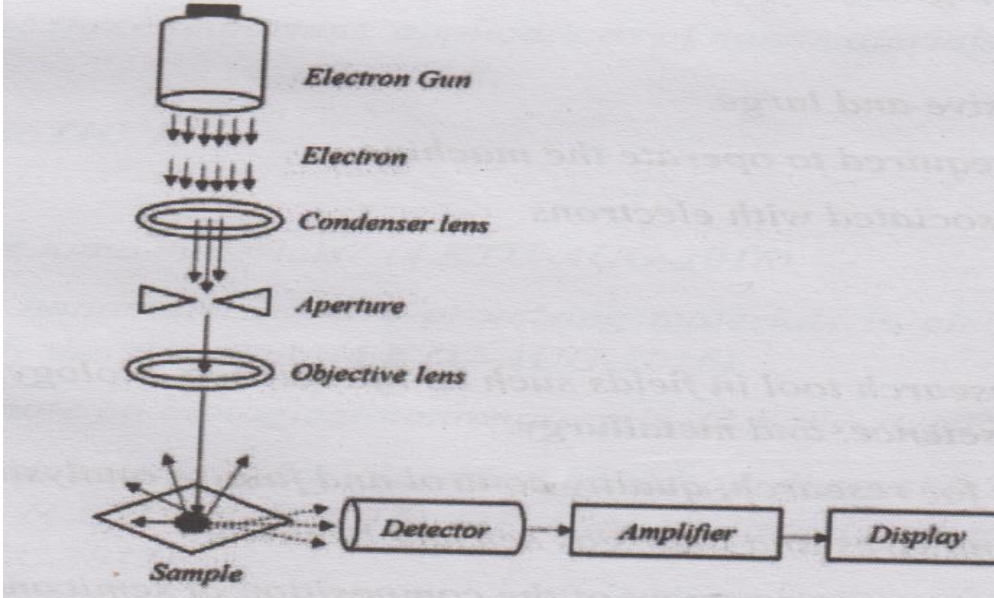
It is used for the surface characterization of materials

SEM can magnify objects upto 1 million times their original size

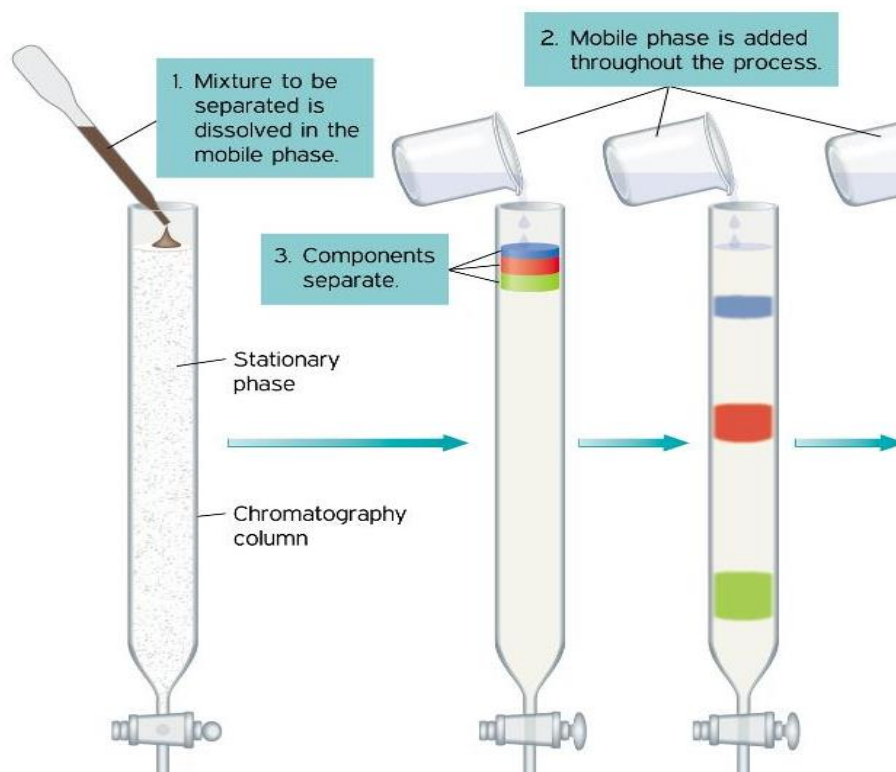
It produces the image of a surface by scanning the surface with a high energy and focused beam of electrons

The electrons interact with atoms on the sample surface producing signals

The accelerated electrons dissipate their kinetic energy in the form of various signals which gives information on topography (surface features), morphology (shape & size of particles), composition crystallographic details etc

		 Element of the instrument (i) Electron Gun- it is the source of electrons- usually a heated tungsten wire (ii) Condenser lens- electrons tend to repel each other & condenser lens compresses the electrons to a narrow beam (iii) Aperture- It controls the diameter of the electron beam (iv) Objective lens – It focuses the electron beam to the sample Sample chamber- Encloses the sample Detector- to detect the signals Amplifier- to amplify the signals Display – To show the SEM image. SEM produces black and white three dimensional images
Q16		a) Describe the principle and steps to be followed in column chromatography b) How are nanomaterials classified on the basis of dimension?
ans		16a) COLUMN (ADSORPTION) CHROMATOGRAPHY Substances with almost similar physical and chemical properties cannot be separated from their mixtures by ordinary methods such as distillation, crystallization etc. But if they have different adsorption tendency towards a particular solid adsorbent this fact can be taken into advantage for their separation Adsorption Chromatography is the method of separation of components present in a mixture of compounds based on the difference in adsorption coefficients of these substances towards a particular adsorbent Process Process involves the following steps Step I : <u>Separation of compounds</u> The mixture of compounds (say A+B+C) is dissolved in a suitable solvent and poured down a column filled with a finely divided adsorbent (such as alumina, silica etc.). The most readily adsorbed substance (say A) will get adsorbed at the top of the adsorbent

column along with smaller amounts of B and C. The less readily adsorbed substances (B & C) penetrate various distances down the column and are adsorbed on the lower layers of the column.



Thus a partial separation of the substances is achieved. The separation can be further improved by passing the same or another pure solvent slowly through the column for sometime. The process of improving the separation is known as development of chromatogram and the solvent used is known as developer. The banded column (A band, B band, C band) of the adsorbed substances is called Chromatogram.

Step II : Identification of compounds

If the components in the mixture are coloured substances, a series of coloured rings / Zones / bands are produced on the column.

If the substances are colourless, the fluorescence of the substance is observed by exposing the column to UV light.

Another method is spraying of suitable chemical reagent on the column when it reacts with the substances and develop colour.

Step III : Recovery and Estimation

In order to recover the substances, the chromatogram is pushed out of the glass tube and various zones are cut at the boundaries. These cut portions are then dissolved separately in suitable solvents and are recovered or estimated. This process of recovery of substances is known as elution and the solvent used for the purpose is known as eluent.

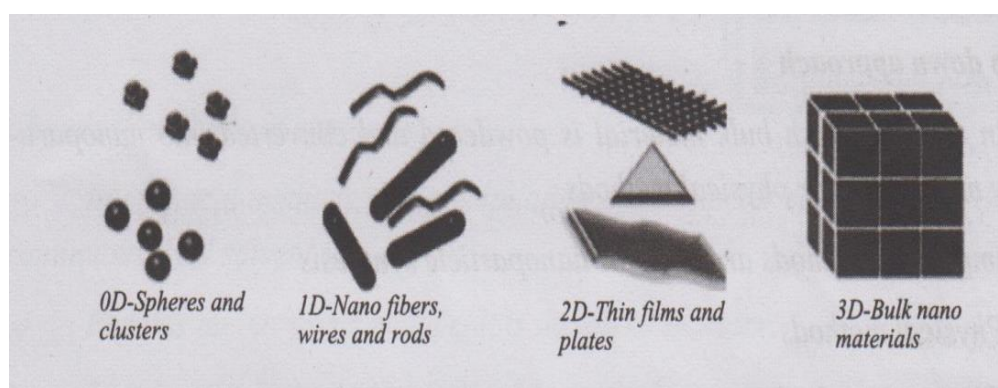
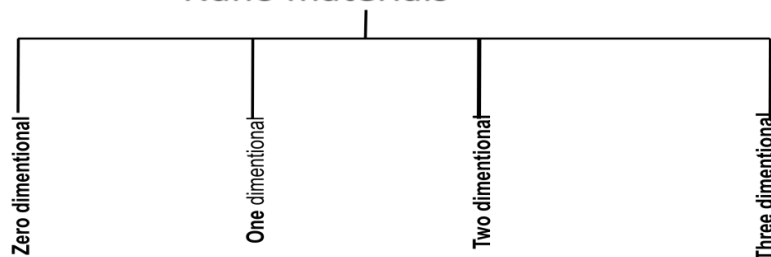
16b)

Nano materials- Classification

There are two classifications for Nanomaterials

■ Classification based on dimensions

Nano materials



Zero- dimensional nanomaterials

For this all dimensions (x, y, z) of the materials are at nanoscale (ie no dimensions are greater than 100 nm)

Eg: nano spheres and nanoclusters

One- dimensional nanomaterials

Here any two dimensions (x, y) are at nanoscale and the third is outside the nanoscale

Eg: nano tubes, nanorods, nanofibres

Two- dimensional nanomaterials

These have at least one dimension (x) at the nanoscale and the other two dimensions are outside the nanoscale

Eg. Nanofilms, nanolayers

Three- dimensional nanomaterials

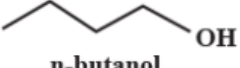
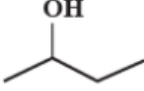
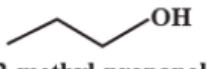
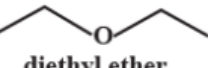
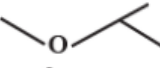
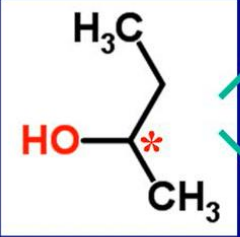
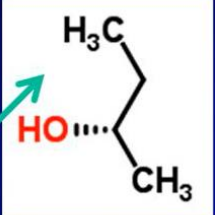
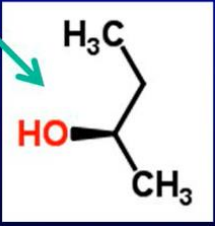
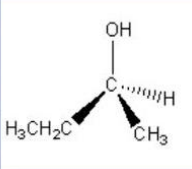
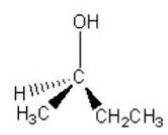
None of the dimensions are within the nanoscale

Eg. Bundle of nanowires / tubes, multi nano layers, cluster of nano particles

Module 4

Q17

a) How many isomers (both structural and stereo) are possible for $C_4H_{10}O$? Draw the structure of each. (8)

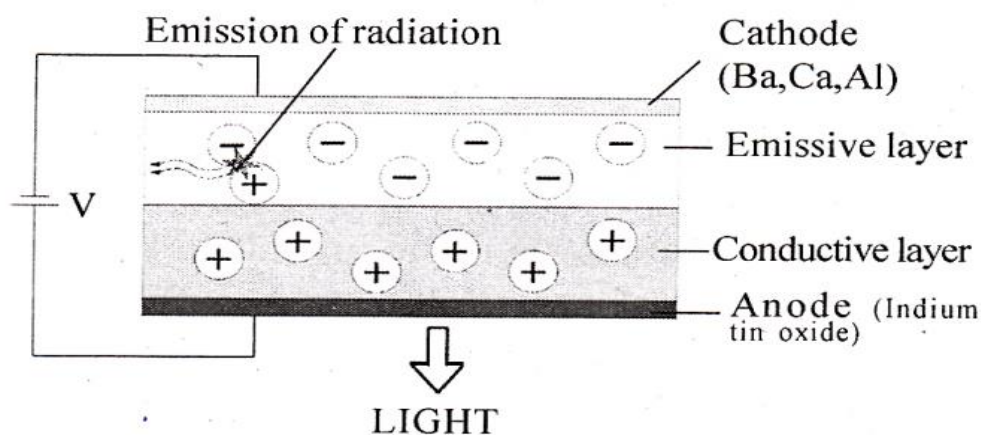
	b) What are OLEDs? Explain the construction and working of OLEDs. (6)
ans	17a) (i)  n-butanol (ii)  butan-2-ol (iii)  2-methyl-propanol (iv)  2-methyl- propane-2-ol (v)  diethyl ether (vi)  1-methoxy propane (vii)  2-methoxy propane Butanol-2 is only having chiral centre. So, 2 optical isomers. So, $C_4H_{10}O$ has 8 isomers including the optical isomers, 7 structural isomers and 1 stereo isomers. 1 chiral center $\rightarrow$ 2 optical isomers  butan-2-ol (2S)-butan-2-ol    

17b)

OLEDs are solid-state devices composed of thin films of organic molecules that emits light when electric current is applied across it (electroluminescence). OLEDs can provide brighter, crisper displays on electronic devices and use less power than conventional light-emitting diodes (LEDs) or liquid crystal displays (LCDs) used today

Working Principle

The battery or power supply of the device containing the OLED applies a voltage across the OLED. A current of electrons flows from the cathode to the anode through the organic layer. i.e. the cathode gives electrons to the emissive layer of organic molecules. The anode removes electrons from the conductive layer on the other side of the organic molecules. This is equivalent to pumping "holes" to the conductive layer

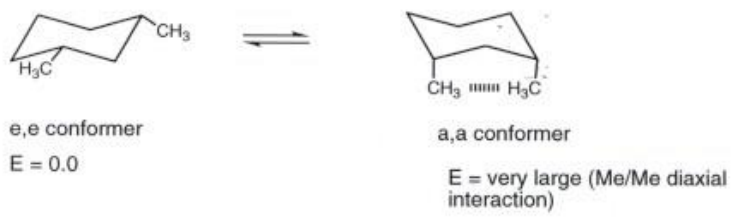


At the boundary between the emissive and the conductive layers, electrons find holes. When an electron finds a hole, the electron fills the hole. As the electrons drop into these holes, they release their extra energy as light. Thus the emissive layer emits light. The color of the light depends on the type of organic molecule in the emissive layer. Manufacturers place several types of organic films on the same OLED to make color displays.

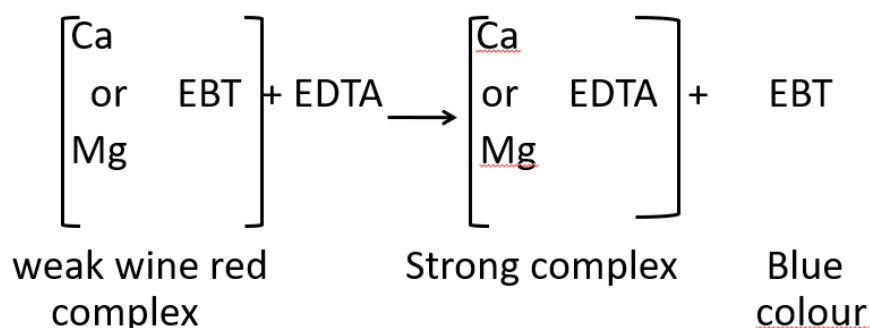
Module 4

Q18

a) Draw the cis and trans isomers of 1, 3-dimethyl cyclohexane. Which will be optically active? Draw all the conformers. Which conformer is more stable and why?(8)

	b) How is polyaniline synthesized? List any two properties and applications. (6)
ans	18a) CIS DIASTEREOMER  Trans Diastereoisomer  Cis 1,3 dimethyl cyclohexane (Diequatorial) is more stable as there is no steric repulsion in this structure and is optically inactive whereas, trans 1,3 dimethyl cyclohexane is less stable due to 1,3 steric repulsion and is optically active. 18b) It is prepared by the oxidative polymerisation of aniline by ammonium persulphate in aqueous HCl medium at lower temperatures. $ n \text{ (Aniline)} + n \text{ O} \longrightarrow \left[\text{C}_6\text{H}_4\text{-NH} \right]_n + n \text{H}_2\text{O} $ Aniline <u>Polyaniline</u>

	Properties High stability in air and solubility in some organic solvents Has wide and controllable range of conductivity Colour change in different oxidation states- used for sensors, chemical sensitivity etc Applications It is used in sensors, Printed circuit boards (PCB), conductive fabrics, smart windows It is used as corrosion resistant material to coat metals
	Module 5
Q19	a) Explain the EDTA method for the estimation of hardness of water with calculation steps.(10) b) Write the procedure for estimating COD of a sample of waste water. (4)
ans	19a) This is a complexometric method of determination of hardness of water by using EDTA (Ethylene Diamine Tetra Acetic acid). In this method, a fixed volume of hard water is titrated against EDTA solution using EBT (Eriochrome Black –T) as the indicator at a pH 8- 10. EDTA- can form strong complex with Ca^{2+} and Mg^{2+} ions present in the hard water. EBT- can form weak complex with Ca^{2+} and Mg^{2+} ions present in the hard water. Before starting the titration when EBT (indicator) is added to the water, Ca^{2+} and Mg^{2+} ions will form a weak complex with the EBT which is wine red in colour. When EDTA is added to this, weak EBT is replaced by EDTA to form strong complex and the indicator is regenerated when colour changes to blue. $ \begin{array}{ccc} \left[\begin{array}{c} \text{Ca}^{2+} \\ \text{or} \\ \text{Mg}^{2+} \end{array} \right] & + \text{EBT} \longrightarrow & \left[\begin{array}{c} \text{Ca} \\ \text{or EBT} \\ \text{Mg} \end{array} \right] \text{Complex} \\ \text{Hard water} & & \text{weak wine red} \\ & & \text{complex} \end{array} $



Step1:Preparation of Standard hard water:

Dissolve accurately weighed X g of CaCO_3 in dil HCl and make upto 1000ml in a standard flask.

Molarity (M1)=Wt per litre/Mol wt = $X/100M=0.0XM$

Step 2: Standardize the EDTA solution (finding out the molarity of EDTA M2)

Standard hard water X EDTA

$$M_1 V_1 = M_2 V_2$$

Let V2 be the volume and M2 be the Molarity of EDTA and V1 be the volume and M1 be the Molarity of std hard water.

$$M_2 = M_1 V_1 / V_2$$

Step 3: Determination of Total Hardness of unknown Water Sample)

To find out the total hardness present in the given sample hard water, V3 ml of unknown water sample is titrated against EDTA as above. Let the volume of EDTA used is V4 ml.

Unknown water sample X EDTA

$$M_3 V_3 = M_4 V_4$$

$$M_4 = M_2$$

$$M_3 = M_4 V_4 / V_3$$

$$\text{Total hardness} = M_3 \times 10^5 \text{ ppm}$$

Step 4: Determination of permanent hardness:

To determine permanent hardness, temporary hardness is to be eliminated and the remaining hardness will be permanent hardness.

For this, pipette out a known volume of hard water sample (V5). Boiled for about 20

	minutes during which temporary hardness causing salts will precipitate out as CaCO_3 and $\text{Mg}(\text{OH})_2$. This is removed by filtration and the filtrate (contains only permanent hardness). To the washing add 10ml of buffer and few drops of EBT indicator. Volume of EDTA consumed = V_6 Standard water sample (boiled) X EDTA $M_5 V_5 = M_6 V_6 \quad M_6 = M_2$ $M_5 = M_6 V_6 / V_5$ Permanent hardness = $M_5 \times 10^5$ ppm Temporary hardness = Total hardness – Permanent hardness 19b) A 250 ml portion of the sewage is refluxed with a known excess of standard potassium dichromate and dilute sulphuric acid in presence of a little Ag_2SO_4 as catalyst for a period of 1.5 hours. The unreacted $\text{K}_2\text{Cr}_2\text{O}_7$ is then determined by titrating against a standard Mohr's salt (Ferrous ammonium sulphate) solution The same test is repeated by using pure water instead of sewage water as a blank test The COD is then Calculated as $\text{COD in mg / litre} = \frac{(V_2 - V_1) N \times 8 \times 1000}{X}$ Where V_2 and V_1 are the volumes of Mohr's salt solution used by the blank sample and the test sample respectively. N is the normality of Mohr's salt solution X is the sewage sample volume taken for the test
	Module 5
Q20	a) Discuss the action of chlorine as a disinfectant. How is it applied? What is break point chlorination? Write any two advantages of breakpoint chlorination. (10) b) A pure water sample is added with 90 mg carbohydrate (CH_2O) per litre find the maximum BOD possible for the water sample. (4)

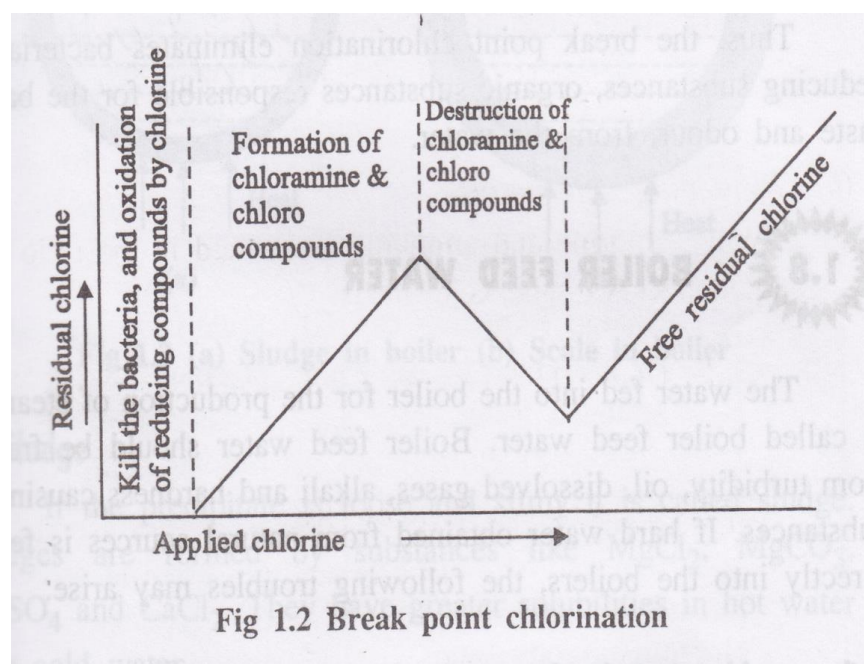
Ans

20a) The process of adding chlorine to water is called chlorination. On adding bleaching powder to water, it produces hypochlorous acid (HOCl). HOCl is a powerful germicide

$$\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$$

$\text{HOCl} + \text{Bacteria} \rightarrow \text{Dead bacteria}$

Even after the purification process, water may contain small amounts of impurities like bacteria, organic impurities, reducing substances (Fe^{2+} , H_2S etc), free ammonia etc. When chlorine is added to water directly.



Stage I: No residual chlorine in the sample. All chlorine added is consumed for oxidation and destruction.

Stage II: The amount of residual chlorine increases with increasing dose of chlorine. Some organic compounds defy oxidation at lower chlorine concentrations. chloro-organics and chloramines are formed.

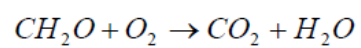
Stage III: Amount of residual chlorine decreases with increase in applied chlorine dose. Here the chlorine added is consumed for oxidation of chloro-organics and chloramines etc.

Complete oxidation and disinfection occur at break point of chlorination.

Stage IV: The applied chlorine dose is not consumed so left us residual chlorine.

Advantages of break point of Chlorination

1. Complete destruction of organic compound.
2. Completely destroys all disease causing bacteria.
3. Prevents the growth of any weeds in water.
4. Removes colour, odour and taste from water

20b)

30g 32g

$$\text{Wt. of oxygen} = \frac{32 \times 90}{30} = 96 \text{ mg / L } \text{ or } \text{ ppm}$$

